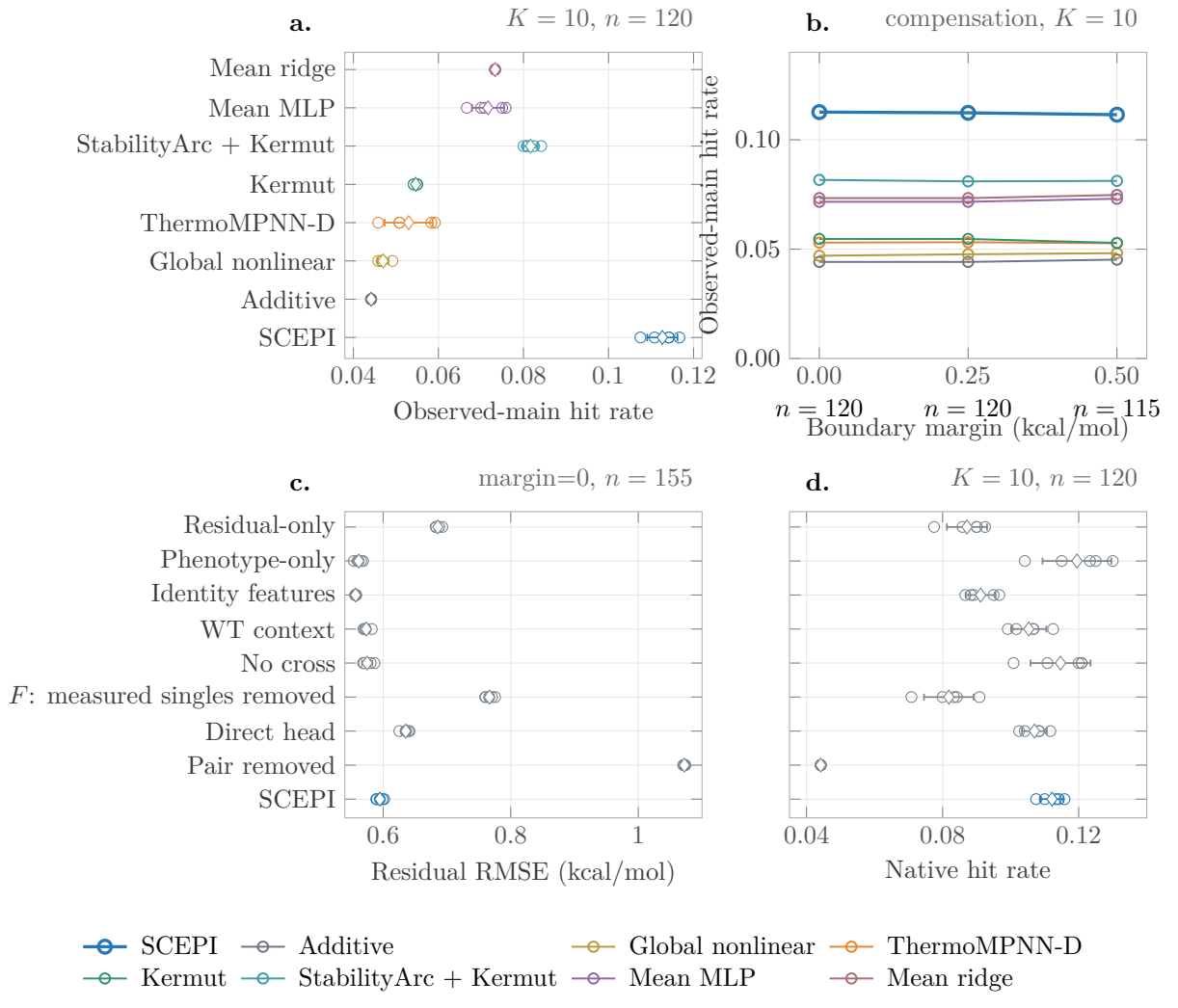
a.
$K = 10$, $n = 120$
Mean ridge
Mean MLP
StabilityArc + Kermut
Kermut
ThermoMPNN-D
Global nonlinear
Additive
SCEPI
0.04
0.06
0.08
0.1
0.12
Observed-main hit rate
Observed-main hit rate
b.
compensation, $K = 10$
0.10
0.05
0.00
0.00
0.25
0.50
$n = 120$
$n = 120$
$n = 115$
Boundary margin (kcal/mol)
c.
margin=0, $n = 155$
Residual-only
Phenotype-only
Identity features
WT context
No cross
F: measured singles removed
Direct head
Pair removed
SCEPI
0.6
0.8
1
Residual RMSE (kcal/mol)
d.
$K = 10$, $n = 120$
0.04
0.08
0.12
Native hit rate
SCEPI
Additive
Global nonlinear
ThermoMPNN-D
Kermut
StabilityArc + Kermut
Mean MLP
Mean ridge